Ex No: 01

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# STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

Aim:-

To study the various network commands used in Linux and windows.

(i) Code:-

```
arp -a
```

Output:-

```
(172.16.11.109) at 00-15:5d:ca:23:54
(ether) on eth0
```

(ii) Code:-

```
hostname
```

Output:-

~~localhost.localdomain~~ LAPTOP-UOAOHQHU

(iii) Code:-

```
netstat -r
```

Output:-

Kernel IP routing table

| Destination | Gateway | Genmask | Flags | MSS | Window |
|---|---|---|---|---|---|
| default | LAPTOP-UAOW | 0.0.0.0 | UG | 0 | 0 |
| 172.17.16.0 | 0.0.0.0 | 255.255.240.0 | U | 0 | 0 |

(iv) Code :-

nslookup www.google.com

Output :-

Server : 10.225.255.254

Address : 10.255.255.254#53

Non-authoritative answer :

Name : www.google.com

Address : 142.251.179.119

(v) Code :-

~~route~~ route

Output :-

Kernal IP routing table

| Destination | Gateway | Genmask | Flags | Metric |
|---|---|---|---|---|
| 172.17.160.0 | 0.0.0.0 | 255.255.240.0 | U | 0 |

(vi) Code :-

ip address show

Output :-

link/loopback 00:00:00:00 brd 00:00:00

inet 127.0.0.1/8 scope host lo

valid_lft forever preferred_lft forever

inet6 ::1/128 scope host

valid_lft forever preferred_lft forever

(vii) Code:-

ifconfig

Output:-

eth0: flags = 4163 <UP, BROADCAST, RUNNING

- inet 172.17.172.190 netmask 255.255.240

inet fe80::215:5dff:fef6:2669

ether 00:1f:f3:ff:26:e9 txqueuelen 1000

RX packets 657 bytes 81452 (81.4 KB)

(viii) Code:-

mtr

Output:-

Hostname: localhost

| Loss | Snt | Last | Avg | Best | Wrst | StDev |
|---|---|---|---|---|---|---|
| 0.0% | 42 | 0 | 0 | 0 | 0 | 0.0 |

(ix) Code:-

tcpdump -D

Output:-

1. eth0 [Up, Running]
2. lo [Up, Running, Loopback]
3. any (Pseudo-device that captures on all interfaces)
4. bluetooth-monitor (Bluetooth Linux)

(x) Code:-

```
nbtstat -s
```

Output:-

```
Resolved by broadcast = 0
Resolved by Name server = 0

Registered by broadcast = 4
Resolved by Name server = 0
```

(xi) Code:-

```
pathping -4 172.16.97.153
```

Output

```
LAPTOP-00A0HQHU [172.16.11.32]
1  172.16.8.1

Hop  RTT    Lost/Sent=Pct    Address
1    0ms    0/100=0%         LAPTOP-00A0HQHU
2    97ms   4/100=4%         172.16.8.1
```

Result:-

This the network commands using Linux and Windows are successfully executed.